# Smart Multi-Sensor Wearable for Real-time Elderly Health Anomaly and Fall Detection

R26-IT-108

Project Proposal Report

Dissanayaka D.M.S.K. - IT22291196

B.Sc. (Hons) in Information Technology Specialization in Information Technology (IT)

Department of Information Technology

Sri Lanka Institute of Information

Technology Sri Lanka

2026 March

# Smart Multi-Sensor Wearable for Real-time Elderly Health Anomaly and Fall Detection (Respiratory & Thermal Warning System)

R26-IT-108

Project Proposal Report

B.Sc. (Hons) in Information Technology Specialization in Information Technology (IT)

Department of Information Technology

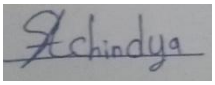
Sri Lanka Institute of Information

Technology Sri Lanka

2026 March

# DECLARATION

I declare that this proposal is my own work and does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by another person except where due acknowledgement is made in the text.

| **Name** | **Student ID** | **Signature** |
|---|---|---|
| Dissanayaka D.M.S.K | IT22291196 |  |

The above candidate is carrying out this research for the undergraduate dissertation under my supervision.

**Supervisor:** Mr. Nelum Amarasena

**Date:** 12.03.2026

**Co-Supervisor:** Samadhi Rathnayake

**Date:** 12.03.2026

# ACKNOWLEDGEMENT

I am pleased to say that I greatly appreciate all the help that people have given to me in the preparation of this research proposal. To begin with I should like to thank my supervisor who guided me in the right direction, fostering me in my study development, and giving me constructive suggestions along the way. Their guidance and academic assistance came in very handy to develop this research in a purposeful and structured way.

I also owe my debt to the lecturers and academic staff of the Sri Lanka Institute of Information Technology whose knowledge, guidance and learning opportunities they have availed to me through the course of my studies. Their assistance has played a major role in the successful development of this research work.

I also wish to offer my sincere thanks to group members and colleagues since they were cooperative, shared ideas, and encouraged me through numerous discussions concerning this project. They have assisted in enhancing the quality of this proposal.

Secondly, I would like to thank my family and friends who were very patient, understanding, and always encouraged me during this time. Their moral encouragement has been very strong to do this work.

Lastly, I wish to thank all authors, scholars, and organizations for the published work and resources of which informed the literature and knowledge base of this study.

# ABSTRACT

The rising trend of ageing has generated a great demand for low-cost as well as continuous healthcare monitoring solutions that can be used to assist the aged people in the non-hospital settings. Silent respiratory deterioration and oxygen desaturation, as well as temperature abnormalities, especially vulnerable to older adults, can be signs of infection, inflammation, or other severe physiological inefficiency. It is important that such conditions be identified early since their later identification may lead to a higher risk of hospitalization and serious health consequences. To address this challenge, this study will present a wearable Respiratory and Thermal Warning System that will help track the main physiological variables in real-time and provide early warning about deviant situations.

The proposed system targets the constant monitoring of the oxygen saturation levels (SpO 2), heart rate and skin temperature using an integrated wearable architecture. The system is equipped with built-in processing to obtain, pre-process and analyse physiological data right on-board to provide low-latency response and enhance privacy. Furthermore, the suggested method uses individualized baseline ranges and abnormality detection logic to detect deviations that can be related to respiratory or thermal threat. In case of unsafe conditions, the device sends real-time alerts to the user and other caregivers connected to the device.

This study is important as it discusses the drawbacks of the current monitoring systems, most of which employ single parameter sensing, costly infrastructure or cloud reliant processing. The anticipated final product is a prototype that is low-cost and preserves privacy to be used in elderly residential monitoring and early intervention process. This work is significant in the wearable healthcare innovation since it integrates multi-parameter sensing, embedded intelligence, and real-time warning capabilities into a real-life elderly care system.

# LIST OF ABBREVIATIONS

| Abbreviation | Full Form |
|---|---|
| AI | Artificial Intelligence |
| API | Application Programming Interface |
| BPM | Beats Per Minute |
| ESP32 | Espressif 32-bit Microcontroller |
| HR | Heart Rate |
| IoT | Internet of Things |
| ML | Machine Learning |
| PPG | Photoplethysmography |
| SDG | Sustainable Development Goal |
| $SpO_2$ | Peripheral Capillary Oxygen Saturation |
| UI | User Interface |
| WBS | Work Breakdown Structure |

# Table of Contents

DECLARATION ..... 3
ACKNOWLEDGEMENT ..... 4
ABSTRACT ..... 5
LIST OF ABBREVIATIONS ..... 6
1. Introduction ..... 9
1.1 Background and Context ..... 9
1.2 Literature Review ..... 11
1.3 Research Gap ..... 13
2. Objective ..... 15
2.1 Main Objective ..... 15
2.2 Specific Objectives ..... 15
3. Methodology ..... 17
3.1 Research Approach ..... 17
3.2 General Methodological Framework ..... 17
3.3 System Design Overview ..... 18
3.4 Hardware Components and Sensor Selecting ..... 19
3.5 Data Acquisition Process ..... 19
3.6 Data Preprocessing Techniques ..... 20
3.7 Baseline Modelling and Threshold Definition ..... 21
3.8 Mechanism of Abnormality Detection ..... 21
3.9 Edge Processing and Embedded Intelligence ..... 22
3.10 Process of Generating and Notifying Alerts ..... 22
3.11 Prototype Development ..... 23
3.12 System Evaluation Methods ..... 23
3.13 criteria of performance measure ..... 24
3.14 Ethical and Safety Considerations ..... 24
4. High-Level System Architecture ..... 25
5. User Requirements ..... 28

5.1 Stakeholder Identification ..... 28
5.2 Functional Requirements ..... 29
5.3 Requirement Description Summary ..... 32
5.4 Expected Test Cases ..... 32
6. Commercialization Plan ..... 36
7. Budget and Justification ..... 37
8. Work Breakdown Structure (WBS) diagram ..... 41
9. GANTT CHART ..... 41
10. References List ..... 42

# 1. Introduction

## 1.1 Background and Context

The ageing of population has emerged as one of the key health and social changes of modern times. With the ever-expanding global ageing population, global health care systems are increasingly being challenged to offer sustained, cost-effective, and preventive care in non-clinical settings. Older people are at higher risk of chronic disease, physiological instability, late symptom identification and emergency incidents that happen during solitary life or under minimal monitoring. In these respects, wearable health technologies have become an effective method of assisting in long-term monitoring of vital signs and providing the opportunity to receive timely medical help without staying in a hospital constantly [1], [2]. The developments are particularly applicable to the ageing population where physiological changes are sometimes subtle in nature without a clear deterioration.

Among the numerous risks of negative effects on older adults, respiratory deterioration and abnormal body temperatures are especially material since they are often linked to grave yet initially asymptomatic medical issues. Oxygen saturation decreases can be a sign of poor respiratory performance, infection, cardiac stress, or poor circulation. Similarly, irregular temperature trends can indicate fever, inflammatory, infection, or general thermoregulatory issues. These red flags in the elderly may be harder to discern by symptoms alone as the ageing body may not react as well as in younger adults and the symptoms do not come out as quickly or are mild in the initial stages. Respiratory-related and pulse oximetry are thus instrumental in detecting deterioration in time before it gets serious [6], [7], [9]. Temperature monitoring is also important as it provides physiological background and works better when paired with oxygen saturation and pulse-related signals in the analysis [10], [12].

Early warning is necessary since respiratory compromise may progress and go unnoticed until the situation turns critical. Studies on pulse oximetry and home monitoring have demonstrated that oxygen destruction can take place prior to the manifestation of overt distress, and thus continuous monitoring can be useful in early intervention and escalation of care [7], [9]. In the case of older adults, the delay in identification can result in hospital admission, morbidity, and mortality. A

proactive warning model can thus act as a supplement to the caregiver and health professional to recognize abnormal patterns instead of depending on the late symptoms. This is in line with the larger trend towards reactive healthcare being replaced with preventive, data-driven, and patient-centered care models [2], [10].

Such monitoring has international and domestic significance. Around the world, healthcare systems are finding scalable solutions to support ageing populations, lessen unnecessary hospital loads and achieve better outcomes using remote monitoring technologies [2], [10], [11]. Locally, the requirement is also pertinent in the context where access to regular clinical evaluation can be restricted and where family-based care of the elderly is still widespread. Low-cost wearable systems, which offer continuous monitoring, can enhance home-based care and enhance safety in such environments without having to install complex infrastructure. Wearable devices also have benefits of mobility, convenience and real-time data collection over episodic clinical measurement [1], [12].

Nevertheless, the current monitoring systems exhibit several shortcomings. Most of the commercial and research-based solutions are based on a single physiological variable, or rely on hospital-grade environments, repeat manual interpretation, or cloud-centered processing [1], [2], [10]. Monitoring the respiration process can involve large devices, modelling of indirect estimates or sensor arrangements that are not comfortable in daily wearables [5], [6]. Even there, pulse oximetry, in its very usefulness, has been known to have limitations about interpretation, environmental factors, and measurement accuracy in various conditions [7], [8]. Moreover, a lot of systems are built on general wellness monitoring but not on early warning in high-risk elderly groups. Such problems decrease viability, cost-effectiveness, and agility in everyday life.

To overcome these drawbacks, this paper introduces a wearable Respiratory and Thermal Warning System to monitor the elderly. This system is intended to combine saturation of oxygen, heart rate, and temperature data by means of a compact wearable structure assisted by embedded smarts. The proposed solution will identify the initial signs of abnormal patterns and provide real-time warnings by integrating various physiological indicators and analyzing them close to the point of sensing. Embedded or edge-based processing is also significant in that it may decrease the latency, enhance privacy and facilitate consistent functionality under resource-intensive environments [3], [10]. In principle, the proposal aligns with the Sustainable Development Goal 3 objectives of

healthy lives and well-being at all ages, as well as with the objectives of innovation in smart healthcare, embedded systems, and digital health research. Being a research contribution, it is located at the crossroads of wearable computing, biomedical signal analysis, elderly care technology, and intelligent edge systems [1], [3], [11].

## 1.2 Literature Review

According to recent literature, wearable technologies in healthcare have grown significantly in continuous physiological monitoring. There is a growing adoption of wearable devices to measure vital signs in ambulatory and home settings, which allows moving the episodic assessment to continuous monitoring [1], [2]. Dias and Cunha referred to wearable health devices as a significant technological base of real-time measuring of vital parameters, highlighting their significance in enhancing accessibility and patient-centred care [1]. Likewise, Lu et al. emphasized the healthcare value of wearable devices in general, such as chronic disease management, preventive care, and remote patient monitoring [2]. This research defines the extensive applicability of wearable sensing as an elderly health monitoring framework.

Regarding respiratory monitoring, there are direct and indirect techniques at present. Reviewing the wearable respiration sensors, Hussain et al. demonstrated that respiratory monitoring could be conducted by sensing the movement of the chest, strain-based approaches, airflow sensing, and bio signal-based estimation [5]. Nevertheless, some of these solutions are hard to downsize to small wearable items that can fit into everyday life, especially when the target audience is older people who need to wear the device conveniently and without any excessive load. Charlton et al. additionally examined breathing rate estimation using electrocardiogram and photo plethysmogram signals and showed that respiration could be determined using signals related to cardiac activity in a non-invasive study [6]. This specifically applies to wearable technologies with PPG-based sensing, since it can be used to interpret respiratory data without necessitating respiration-specific systems.

Pulse oximetry has become one of the most popular respiratory status measurements due to the non-invasive estimate of blood oxygen saturation. In addition to the operational limitations and circumstances that influence reliability, Chan et al. described the physiological basis and clinical value of pulse oximetry [7]. Luks and Swenson highlighted its role in home monitoring, particularly where it is necessary to have an early warning of a worsening situation, but they also cautioned that interpretation should be thoughtful and context sensitive [9]. In more recent studies, Shi et al. showed that accuracy in pulse oximetry can be different between skin pigmentation groups, showing that not all users behave identically when using pulse oximetry [8]. These results indicate that oxygen saturation is very helpful but should be used in conjunction with other indicators in developing strong warning systems.

Another area of significance in remote health surveillance is wearable temperature monitoring. Temperature alone does not necessarily describe respiratory decline, but it adds some useful contextual data concerning infection, inflammation, and the presence of physiological stress. Jiang et al. have outlined wearable tele-health systems that combine various sensors, such as temperature sensors, as a way of aiding in the early detection of health decline [10]. Silva and Tavakoli also wrote about wearable biomonitoring patches being a promising development in home-based hospitalization and remote monitoring specifically where the need to identify worsening effects is essential [12]. These experiments suggest that temperature detection becomes more valuable when it is viewed in combination with oxygenation and cardiovascular measurements.

The next significant theme in the literature is the shift towards edge AI and embedded healthcare processing. Ray discussed TinyML as a new paradigm of implementing machine learning models on low-power embedded systems with advantages like real-time execution, less reliance on network connectivity, and more privacy [3]. The features are useful in the healthcare setting as elderly remote monitoring systems frequently need instant interpretation and notification without delays in communication. The significance of integrated, wearable tele-health architectures that could facilitate decentralized monitoring was also observed by Jiang et al. [10]. In anomaly detection, in particular, Sunny et al. surveyed wearable data analysis frameworks and demonstrated that intelligent methods of detection are more efficient at detecting deviation of normal physiological patterns than straightforward threshold-based monitoring [4].

However, the major achievements notwithstanding existing research and products remain weak in several areas. There are numerous systems available, such as single-parameter, generalized fitness wearables, and multifaceted medical systems that do not optimize toward low-cost usage in the elderly home care [1], [2], [10]. Respiratory techniques can be uncomfortable to use, or they can be indirectly estimated with changing accuracy [5], [6]. Contextual limitations can potentially influence pulse oximetry [7], [8], and multi-sensor platforms might continue to be heavily reliant on cloud computing and without individualized interpretation [3], [4]. Thus, the literature will justify the value of a dedicated, wearable, embedded, and multi-parameter alerts mechanism that would be narrowly tailored to signal the early respiratory and thermal risks to elderly users.

## 1.3 Research Gap

Despite the fact wearable healthcare studies have grown considerably, a valuable void exists in the design of low-cost, senior-centered systems capable of delivering early respiratory and thermal warnings by utilizing integrated, real-time, embedded analysis. The worth of wearable monitoring, respiration sensing, pulse oximetry, anomaly detection, and decentralized healthcare architectures have been validated in existing literature [1],[6], [10],[12]. Nevertheless, such contributions are usually disjointed. There is numerous research that is about wearable health tracking in general, numerous studies are more specifically about respiration or oxygen saturation, and a few studies are more specifically about embedded intelligence without considering the needs of older consumers at risk of silent decline. Consequently, the existing literature fails to fully address the requirement of having a practical warning system that integrates these dimensions into one wearable solution.

The absence of a built-in multi-parameter interpretation is one of the key limitations. Existing systems tend to consider oxygen saturation, respiratory estimation, temperature or heart-related measurements as independent signals when they really constitute a bundle of physiological images [5], [6], [10]. However, the respiratory deterioration of ageing people is seldom captured by a single variable. Desaturation of oxygen can be accompanied by minor changes in heart rate, changes in perfusion, or changes in thermal patterns. When they are tracked separately, the warning system can be slow or less efficient. According to the literature, multi-sensor wearable systems

hold promise, but most of the current applications remain generic tele-health systems and not specifically respiratory and thermal abnormality early-warning systems [10], [12].

The second gap is related to customization and situational intelligence. Those systems based on conventional thresholds can detect extreme values, but not in older monitoring as, at this stage, physiological baseline states differ across individuals. Sunny and colleagues stated that the process of anomaly detection in wear data cannot be confined to dumber rules anymore; it should pay attention to the deviations in the regular patterns [4]. Similarly, embedded intelligence checked by TinyML implies that intelligent interpretation may be executed on a limited hardware [3]. Nevertheless, numerous solutions available are still based on static thresholds, generic alert models, or cloud-based analytics which are not optimal when it comes to personalized and real-time risk identification in homes. This renders existing methods inadequate in detecting early, minor deterioration.

The other weakness is practical deployment. According to research on respiratory monitoring, there are numerous approaches that prove technically effective but do not always allow comfortable everyday use by older adults [5], [6]. Commercial wearables also tend to focus on fitness or wellness, as opposed to clinically meaningful warning functions [1], [2]. Hospital-grade systems, conversely, could provide accuracy, but tend to be prohibitively costly, infrastructure-dependent, or obtrusive to constant use at home [10], [12]. Even pulse oximetry alone is critical, but due to its interpretation and cross-conditional variability, a single SpO2 signal cannot be relied upon to provide reliable risk assessment of the elderly [7],[9]. Thus, the current systems are either too limited, too complicated, or not adequately adjusted to the real-life elderly care.

The novelty of the proposed Respiratory and Thermal Warning System is that it explicitly combines the most vital physiological indicators into an embedded wearable architecture that is specifically designed to be used by elderly users. Instead of considering temperature and oxygenation as independent variables, the proposed system views them jointly, along with heart rate-associated behaviour, to detect early abnormal conditions. The system also focuses on edge-based processing, which can facilitate privacy, low latency, and functionality in environments with limited connections [3], [10]. This increases the responsiveness of the solution and makes it more suitable to be used continuously and at home. The input is hence not merely technological but also

application-focused: it will translate proven sensing technologies into a specialized warning system to a vulnerable population user base.

The study is necessary due to the growing need for elderly healthcare to monitor proactively with affordable, wearable, and instruments that can detect early warning signs when things go out of control. The available literature has valuable fragments of the solution yet fails to sufficiently cover the requirement of having respiratory and thermal alarms, real-time embedded processing, multi-parameter detection, and geriatric-friendly usability in a single coherent system [1], [4], [10], [11]. By bridging this gap, the proposed study will contribute to a practical and research-based prototype that enhances the ability to intervene at an early stage of elderly home care.

# 2. Objective

## 2.1 Main Objective

The primary goal of the research is to design, develop, and test a wearable Respiratory and Thermal Warning System that continuously measures blood oxygen saturation (SpO 2), heart rate, and skin-temperature, and generates real-time early warnings against abnormal respiratory and thermal conditions using edge-based processing. This can be quantified by the successful incorporation of sensing modules, running of embedded anomaly detection logic and measuring whether the system can detect abnormal physiological patterns with prompt alert generation under a wearable prototype setting [3], [10].

## 2.2 Specific Objectives

To achieve the main objective, this study is guided by the following specific objectives:

1. **To develop a wearable sensing module** capable of continuously capturing key physiological parameters, namely blood oxygen saturation ($SpO_2$), heart rate, and skin

temperature, in a non-invasive manner suitable for elderly healthcare monitoring [1], [7], [10].

2. **To preprocess and analyse the collected physiological data** in order to detect abnormal respiratory and thermal patterns that may indicate early signs of health deterioration, including oxygen desaturation, unusual pulse-related changes, and temperature irregularities [5], [6], [12].
3. **To establish personalized baseline ranges** for monitored users by identifying normal physiological trends over time, thereby improving the contextual interpretation of deviations and reducing the limitations of rigid universal thresholds in elderly monitoring [4], [9].
4. **To design and implement a real-time warning mechanism** that issues timely alerts when the system identifies potentially unsafe respiratory or thermal conditions, thereby supporting early intervention by caregivers or healthcare personnel [2], [10], [11].
5. **To incorporate edge-based processing within the wearable system** so that physiological data can be analyzed locally with reduced latency, improved privacy, and less dependence on continuous cloud connectivity, in line with current embedded healthcare approaches [3], [10].
6. **To evaluate the feasibility and practical value of the prototype** as a low-cost elderly monitoring solution by assessing its functionality, responsiveness, and suitability for continuous daily use in home-based care settings [1], [11], [12].

# 3. Methodology

## 3.1 Research Approach

Research design is an applied research project since this study seeks to answer a practical healthcare issue by designing and developing a functional technological system. The study is aimed at developing a wearable Respiratory and Thermal Warning System in the elderly, instead of coming out with mere theoretical results. The research, in this case, adheres to the prototype-oriented approach, in which physiological monitoring, data processing, anomaly detection, and real-time alerting are implemented in an embedded wearable platform. This method is correct since wearable health monitoring studies tend to need repeated development, hardware-software design, and practical validation within natural operating environments [1], [2], [10].

The latest trends in wearable sensing and embedded healthcare intelligence also guide the methodological direction. Earlier studies have revealed that the current trend in continuous health monitoring is moving towards inpatient-based observation to portable and home-based monitoring [1], [12]. Meanwhile, low-power embedded intelligence and TinyML have allowed creating possibilities to process physiological data directly on the limited devices, making real-time monitoring more feasible and privacy-conscious [3]. As such, the proposed work integrates wearable sensing, physiological signal processing and on-device decision-making into one applied research framework.

## 3.2 General Methodological Framework

The research methodology of this work comprises 6 key stages, which include problem identification, system design, hardware integration, data acquisition and preprocessing, abnormality detection, and prototype evaluation. The initial stage is to discover the healthcare monitoring problems related to respiratory deterioration and temperature fluctuations in the elderly. The second step translates these requirements into a technical solution by establishing the

sensing architecture, processing flow and warning mechanism. The third stage is aimed at incorporating a chosen sensor and microcontroller into a wearable prototype.

The fourth stage is the continuous measurement of physiological parameters (blood oxygen saturation, heart rate, and skin temperature), which is preprocessed to eliminate noise and erroneous measurements. The fifth step is the analysis of processed data that detects deviant respiratory and thermal patterns based on joint signal behaviour, individual baselines, and warning standards. The last stage is a test of the system against functionality and how responsive the system is in the elderly monitoring. This general architecture aligns with the logic of development employed in wearable healthcare systems and anomaly detection systems described in previous research papers [4], [10], [11].

## 3.3 System Design Overview

The system, as proposed, will be a small wearable unit that collects physiological information on the user continuously and analyzes it in real-time to provide information about possible early warning indicators. Its design has four main layers: sensing, preprocessing, embedded analysis and alerting. The system gathers SpO 2, heart rate and temperature data in the sensing layer using built-in sensors. The raw measurements are filtered and checked in the preprocessing layer and then analyzed. The system perceives the interpreted values in the embedded layer of analysis to ascertain whether the values are out of the ordinary. The system provides warnings in the alerting layer, both as local notification and system communication.

This design is informed by the fact that one parameter monitoring is not usually adequate to provide reliable elderly care. It has been demonstrated that physiological degradation is better explained by mixtures of related measures instead of single ones [4], [10]. Based on this, the design of the system should integrate indicators of temperature and oxygenation into a single warning model. This facilitates the detection of abnormal conditions at an earlier stage and enhances the utility of the device in domestic-based care environments.

## 3.4 Hardware Components and Sensor Selecting

The hardware design focuses on a low-power embedded system that can be used in wearables. The ESP32 microcontroller is chosen as the main processing unit, due to its computing power, wireless connectivity, and its adaptability to edge computing embedded systems. Its features allow it to be used in the implementation of lightweight physiology data analysis and remain practical in a wearable prototype. Embedded processing use corresponds to the latest TinyML trends and recent trends in decentralized healthcare, which focus on local-intelligence and low-response-time [3], [10].

In the case of physiological sensing, the system employs a sensor based on photoplethysmography to measure blood oxygen saturation and heart rate, and a non-contact or infrared temperature sensing sensor to measure temperature-related changes. This choice is supported by the clinical importance and prevalence of pulse oximetry in the non-invasive respiratory monitoring [7], [9]. Research has indicated that oxygen saturation levels and pulse-related measures can give significant information regarding respiratory discomfort particularly in continuous measurements [6], [7]. The temperature sensing is also added because the presence of thermal abnormalities can signify infection, inflammation, or other related physiological instability that can further enhance the consistency of the warning system when combined with oxygenation data [10], [12].

Other supporting parts might comprise of power regulation parts, rechargeable battery support, and communication interfaces needed to give local alerts or mobile connectivity. The ultimate hardware platform is supposed to be compact, lightweight, and fit in with the everyday lifestyle of the aged citizens.

## 3.5 Data Acquisition Process

The machine constantly takes three primary physiological measurements, blood oxygen saturation, heart rate, and skin temperature. The wearable sensing module captures these values at pre-established intervals and transmits them in their entirety to the embedded controller to be processed. Constant attention is significant since physiological degeneration can take its course

over time and an individual point of measurement might not reveal where an abnormal state is developing [1], [2].

The process of data acquisition is aimed at non-invasive and user-friendly monitoring. The reason behind pulse oximetry-based measurement is that pulse oximetry offers a viable means to estimate oxygen saturation without having to perform invasive operations, whereas heart rate can be obtained based on the same optical sensing process [7]. Temperature measurements are also taken at the same time to give additional physiological background. This system records consecutive readings with time thus allowing not only instant cheques but also the monitoring of short-term trends and an abnormality of behaviour. The significance of this temporal dimension lies in the fact that the detection of abnormalities in wearable devices commonly relies on patterns, but not a specific set of values [4].

## 3.6 Data Preprocessing Techniques

Physiological data obtained using wearable sensors may be filled with motion-induced noise, unreliable measurements, gaps in value sets, and temporary distortions. Thus, before abnormality detection is done, preprocessing must be done first. Preprocessing in this study involves cleaning of signals, filtering of outliers, checking of validity and filtering of measurement variation. These measures enhance the accuracy of the data and decrease the failure of attention when the sensor artefacts occur or the temporary disturbances.

In pulse related and oxygen saturation measurements, preprocessing can stabilize measurements that are due to motion, changes of contact, or optical interference. This is necessary particularly due to the sensitivity of pulse oximetry to environmental and physiological conditions [7], [8]. Implausible values in temperature are also checked and normalized to ensure consistency in interpretation over time. In cases where readings are invalid or incomplete, the system uses handling rules of temporarily rejecting or replacing based on adjacent valid values. These steps align with wearable anomaly detection principles, in which the quality of input has a significant effect on the detection performance [4]. The processed data is then transformed into input to be analyzed to make a baseline comparison and identify abnormalities.

## 3.7 Baseline Modelling and Threshold Definition

One of the main methodological characteristics of the proposed system is the development of user specific physiological baselines. The system does not rely solely on strict universal thresholds but instead it uses a window period to approximate the normal range of each user regarding patterns in SpO 2, heart rate, and temperature. This is significant since elderly people might have varying physiological conditions during rest and one value might be good in one person but a sign of deterioration in another [4], [9].

There are two ways in which threshold definitions are used. The first step is to put general safety thresholds and determine the specific, evidently dangerous values that need instant warning. Second, deviations are detected using personalized baseline ranges that do not necessarily have to be clinically extreme but is still a significant change in the normal state of the user. The combination of the two methodologies is justified by the body of literature on the concept of anomaly detection, which indicates that a pattern-based deviation detection technique can be more sensitive to detecting anomalies than purely fixed-rule systems [4]. Baseline modelling is especially applicable when dealing with elderly care because a gradual decline might be a more informative indicator than a single breach of a severe threshold.

## 3.8 Mechanism of Abnormality Detection

The mechanism of the abnormality detection takes the preprocessed physiological inputs and detects the respiratory and thermal warning conditions. The detection logic is of combined interpretation as opposed to the relying on isolated signal analysis. Indicators of a greater probability of deterioration than any single indicator, e.g., reduced oxygen saturation, atypical pulse-related variation, atypical temperature elevation or instability could be combined. This is multi-parameter reasoning that is backed by the previous tele-health and wearable monitoring studies [10], [12].

The detection mechanism can also include rule-based logic along with lightweight and embedded-executable anomaly detection mechanisms. Rule-based detection offers interpretability and safety

of known abnormal conditions and anomaly-oriented analysis assists in identifying unusual physiological patterns when compared to the norm [4]. This will enable the system to identify the immediate violations of the threshold as well as minor deviations that arise with time. The result of this step is a risk decision, which takes into consideration whether the user is at normal, cautionary or a warning condition.

## 3.9 Edge Processing and Embedded Intelligence

One of the key methodological aspects of the proposed system is the processing with the help of edges. The system does not send all the raw sensor data to a remote server and analyses it but instead uses the embedded device to perform the primary interpretive functions. This architecture minimizes latency, reinforces privacy, and enhances operational autonomy in the environments, where connectivity might be unavailable occasionally [3], [10].

Literature provides solid support on the suitability of edge intelligence in this system. Recent studies in the field of tinyML indicate that small machine learning and decision models have now come to be able to be ascended to resource-constrained devices and still implement inference capability in a practical capacity [3]. This is particularly applicable to elderly healthcare since the warning systems should always be operable and react promptly to any changes in physiology. Embedded processing also means that the system is not forced to always rely on clouds, and this can enhance reliability and can help the system to be more appropriate to the reality of a domestic environment. As such, embedded intelligence is integrated within this study as a technical but also a healthcare-based design philosophy.

## 3.10 Process of Generating and Notifying Alerts

When an abnormal respiratory or thermal condition has been detected, the system triggers its warning mechanism. Alerts are set to be emitted in real time such that the user, caregiver or interconnected monitoring interface might be informed immediately. Local feedback may be

involved in the warning process, like vibration or on-device notification, and communication to an external mobile or monitoring application.

The alerting plan is organized based on severity. Minor deviations can cause notifications operating on the caution level whereas continual or severe anomalies will result in warnings operating on a higher level. This progressive method is suitable since the persistent wearable monitoring might result in various physiological deviations, with not all of them warranting the identical reaction. As noted, research in remote monitoring argues that timely and interpretable alerts are the key to the practical use of tele-health systems [2], [10], [11]. In this regard, the proposed approach will make sure that the generation of alerts is both timely and clinically significant.

## 3.11 Prototype Development

The prototype is built with software and hardware modules that are integrated on a recursive basis. Hardware assembly comprises the connection of the sensing elements with the ESP32 platform, power support configuration, and wearable configuration to be used in physiological measurements. At software level, firmware is written to control sensor readings, preprocessing, baseline estimation, abnormality detection and sending an alert.

Prototype development is done in phases to enable testing and refinement of individual modules before complete system integration. The first one is the verification of individual sensor functionality. Then, combined data acquisition is tested. Then, preprocessing and detection programmes are integrated into the system. Lastly the complete prototype is tested as a functional wearable device. This gradual development procedure is representative of the usual practice in the implementation of wearable systems and minimizes the chances of integration failures [1], [10], [12].

## 3.12 System Evaluation Methods

Functional and performance-oriented tests are used to evaluate the system. Functional testing ensures that the prototype can accurately measure SpO 2, heart rate, and temperature data, perform

embedded analysis, and produce alerts when necessary. The performance-based evaluation looks at the responsiveness, stability and viability of the wearable device in operation.

The assessment also involves the ability to detect abnormality. This involves investigating the existence of physiological deviations (simulated or controlled) that evoke the desired warning outputs. Also, repeated readings and time-variance of system behaviour are evaluated. These assessment tools align with the focus of wearable healthcare studies on usability, responsiveness and meaningful clinical utility as opposed to theoretical algorithmic performance alone [1], [11], [12].

## 3.13 criteria of performance measure

Evaluation of prototype performance is conducted against several criteria. The former is the detection effectiveness, which is the capacity of the system to detect respiratory and thermal abnormal patterns accurately and reliably. The second one is response time, which is defined as the speed at which the system sends warnings once abnormal conditions have been detected. The third is signal stability that indicates stability of sensor readings during constant wearable use.

Other requirements are privacy-saving features with local processing, energy-saving to be used in practical wearable applications, and feasibility in general to implement in elderly care. These criteria are essential since the success of the system is not only determined by the system itself being operable, but also the ability of the system to be employed, timely and suitable in real world monitoring [3], [10], [11].

## 3.14 Ethical and Safety Considerations

This paper pays close consideration to ethical and safety issues related to wearable health monitoring. Because the system deals with physiological data, privacy and sensitive management of information involving health-related data matters. Privacy preservation is also facilitated using

edge processing that reduces unnecessary transmission of raw data [3]. Moreover, the wearable design should be safe, non-invasive, and comfortable enough to be used by the target population.

It is restricted to prototype monitoring and warning support. It is not meant to serve as an alternative to professional diagnosis or emergency medical systems. Thus, the device generates alerts, but these alerts are used as early notices and not conclusive clinical judgement. This difference is significant as pulse oximetry and wearable sensing are useful, but they have been associated with known limitations and should be construed with caution [7],[9]. The ethical methodology in this regard then needs to be transparent regarding capability of the system, responsible use of data and safe use of the device.

# 4. High-Level System Architecture

The Respiratory and Thermal Warning System is a proposed wearable device based on edge-enabled health monitoring as an extension of the larger elderly healthcare framework. Its primary objective is to constantly monitor vital physiological signals and locally analyse them and provide early warning when respiratory or temperature deviations are detected. At the top, the architecture comprises five logical layers, the sensing layer, embedded processing layer, decision and alerts layer, communication layer, and the external monitoring layer. This stratified design facilitates continual observing of physiological activity whilst ensuring low latency, privacy and usability by elderly patients in their residential settings [1], [3], [10].

The wearable device captures physiological data that is appropriate to measure respiratory and thermal conditions at the sensing layer. Blood oxygen saturation and heart rate are monitored by an optical sensing module and temperature is monitored by a specific thermal sensor. These measurements are selected since combined information on oxygenation, pulse-related behaviour, and temperature variation gives a more significant picture of potential respiratory deterioration than that of any individual parameter alone [6], [7], [10]. This is the research element of the current study and in the general team system it is the module that deals with early respiratory and thermal risk identification.

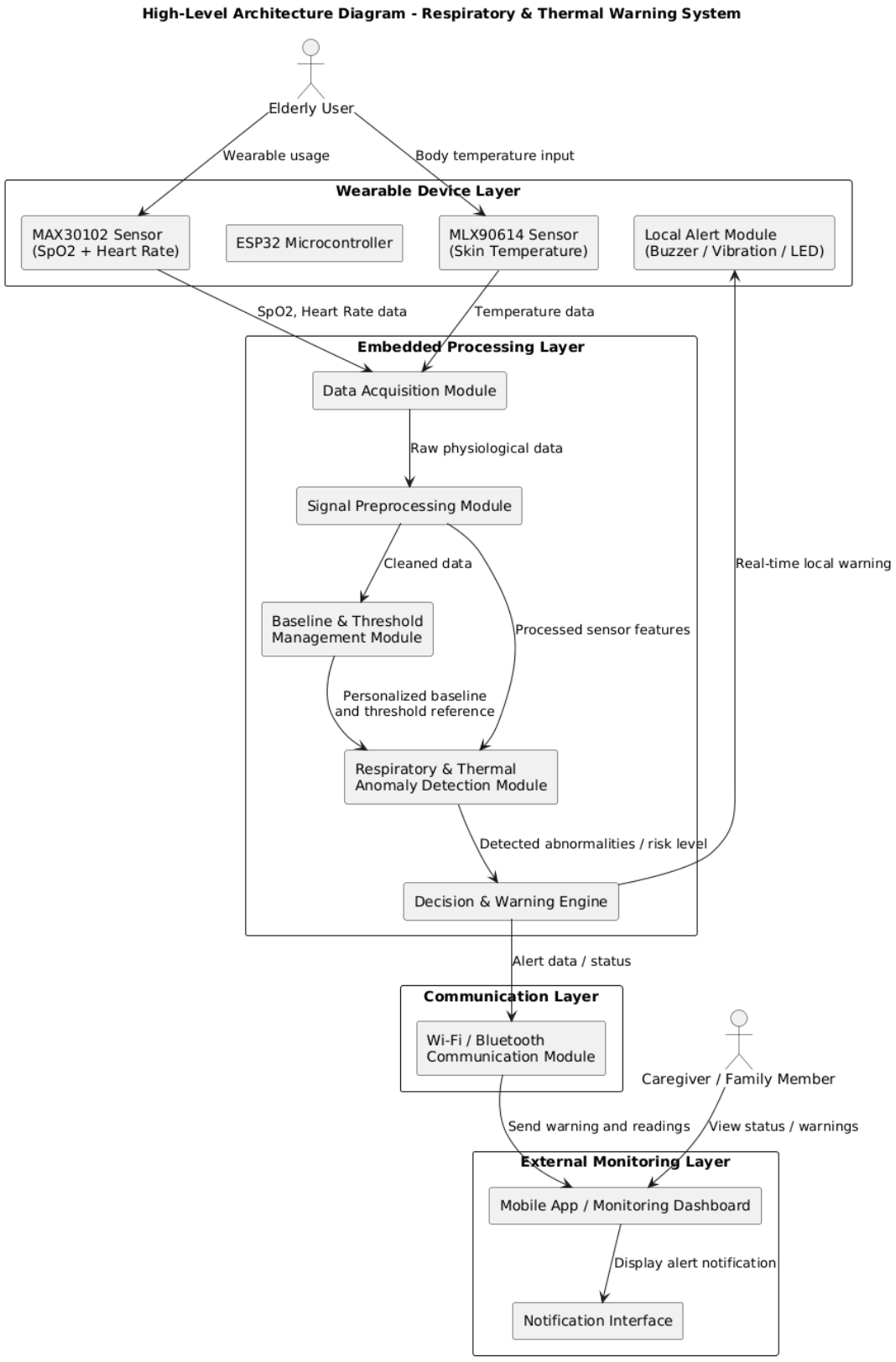



***Diagram 1- High level architecture diagram***

The proposed architecture is centered on the embedded processing layer. Once sensor readings have been received, the data acquisition module forwards raw values to preprocessing functions that process signal cleaning, removal of invalid-readings and fundamental stabilization. After preprocessing, the data are tested by both general thresholding rules and individualized baseline ranges to recognize abnormal respiratory or thermal configurations. The significance of this local processing method is that wearable healthcare systems demand quick analysis of physiological transformations, and embedded intelligence can handle such examination without the delays incurred by cloud-based systems [3], [4]. On-device analysis usage also points to the current trends in wearable and tele-health architectures that are characterized by responsiveness and decentralized decision-making [10], [12].

The analyzed results are sent to the decision and alert layer, which identifies the user as being in normal, cautionary, or warning state. In case of an abnormal condition being detected, a real-time alert mechanism is triggered by the system. This can be local output by vibration, buzzer, or other means, and the user can get an immediate warning. Moreover, the identified condition can be directed to an outermost application layer to enable caregivers or stakeholders to detect the status and take the necessary measures. Such an alert-focused design is in line with the goal of early intervention, which is particularly relevant in the context of elderly monitoring, where silent degradation can otherwise go unnoticed [2], [9], [11].

Communication layer links the wearable system with outside monitoring services that include a mobile application or dashboard. This layer is used in transmitting warning messages, summary of physiological status, and event notifications. The architecture is such that primary detection is done locally on the device though. It is a major benefit of security and privacy since sensitive physiological information does not have to be constantly sent to be analyzed remotely. Processing of edges thus minimises exposure of personal health information and enhances confidentiality through the minimization of redundant information sharing [3], [10]. External communication of only essential results, warnings, or summed up reading should be done when necessary.

In terms of data flow, architecture has a definite direction. To begin with, wearable sensors record physiological signals on the user. Second, the raw readings are sent to the data acquisition and preprocessing modules. Third, the evaluation of cleaned data is conducted by the logic of baseline comparison and abnormality detection. Fourth, the decision engine identifies the physiological

condition and decides whether a warning is to be issued. Lastly, the warning is issued locally and, should it be turned on, reported to the external monitoring interface. This chain of command makes sure that the system can provide real-time monitoring in a well-organized and efficient way.

A key architectural element in this research component is security. Since the system involves health related data, it needs privacy preserving design. The implementation of edge-based processing limits the need to have continuous cloud connectivity, minimises the risks of data transmission, and facilitates the safer treatment of physiological data [3]. Moreover, a controlled communicative device may be employed in such a way that only the alert that is necessary and the output values that are selected are disseminated. This enhances user confidence and makes the architecture more healthcare ready.

The high-level design also considers scalability. The architecture is modular, i.e., it is possible to add other sensing modules, analytics functions or communication services in the future without re-architecting the whole system. As an illustration, the same embedded system might subsequently be expanded to facilitate a more sophisticated anomaly detection, the connections with the wider system of smart wearables, or more extensive tele-health monitoring capabilities [10], [11]. As such, the suggested high-level architecture is capable not only of supporting the short-term goals of respiratory and thermal warning, but also a scalable platform upon which future upgrades and integration with the entire system can be achieved.

# 5. User Requirements

This section defines the key stakeholders and functional requirements of the proposed Respiratory and Thermal Warning System. The requirements are derived from the needs of elderly monitoring, early warning support, and integration with the overall smart wearable healthcare platform.

## 5.1 Stakeholder Identification

*Table 5.1: Stakeholder Identification for the Respiratory and Thermal Warning System*

| Stakeholder | Description | Role in the System | Main Needs |
|---|---|---|---|
| Elderly Users | Primary end users who wear the device for continuous health monitoring | Generate physiological data through daily wearable use and receive direct alerts | Comfortable monitoring, real-time warnings, ease of use, safety |
| Caregivers / Family Members | Relatives or responsible people who support the elderly user | Receive alerts and monitor abnormal conditions for timely response | Timely notifications, clear warning information, remote visibility |
| Healthcare Providers | Medical professionals who may review warning outcomes or physiological trends | Interpret warning-related information for further assessment or intervention | Reliable physiological data, meaningful alerts, clinical relevance |

*Table 5.2: Stakeholder Expectations*

| Stakeholder | Expected Benefit from the System |
|---|---|
| Elderly Users | Early identification of respiratory or thermal abnormalities without continuous hospital supervision |
| Caregivers / Family Members | Ability to respond quickly when the elderly user shows signs of deterioration |
| Healthcare Providers | Access to useful monitoring information that can support preventive care and follow-up decisions |

## 5.2 Functional Requirements

*Table 5.3: Functional Requirements of the Respiratory and Thermal Warning System*

| Requirement ID | Functional Requirement | Description |
|---|---|---|
| FR-01 | Respiratory Data Collection | The system shall continuously collect physiological data related to respiratory condition, including $SpO_2$ and heart rate, from the wearable sensing unit. |
| FR-02 | Temperature Sensing | The system shall continuously measure the user's skin/body temperature using the integrated thermal sensor. |
| FR-03 | Real-Time Data Acquisition | The system captures and transmits sensor readings to the embedded processing module at predefined intervals for immediate analysis. |
| FR-04 | Data Preprocessing | The system shall clean and validate raw sensor readings by reducing noise, filtering invalid values, and stabilizing measurements before analysis. |
| FR-05 | Personalized Baseline Establishment | The system shall establish a normal physiological baseline for the user based on collected historical readings. |
| FR-06 | Abnormality Detection | The system shall identify abnormal respiratory and thermal patterns by analyzing $SpO_2$, heart rate, and temperature data against thresholds and baseline values. |
| FR-07 | Risk Classification | The system shall classify the detected state into normal, caution, or warning conditions based on the severity of abnormal patterns. |
| FR-08 | Real-Time Alert Generation | The system shall generate immediate alerts when abnormal respiratory or thermal conditions are detected. |
| FR-09 | Local Warning Output | The system shall notify the elderly user through local alert mechanisms such as vibration, buzzer, or visual indication. |

| | | |
|---|---|---|
| FR-10 | External Notification Support | The system shall send warning information to the connected mobile app or monitoring interface for caregivers or relevant users. |
| FR-11 | Continuous Monitoring Support | The system shall operate continuously during wearable use to support uninterrupted physiological monitoring. |
| FR-12 | Communication with Main Team System | The system should exchange warning status and monitored data with the overall team healthcare platform for integrated monitoring and response. |

***Table 5.4: Functional Requirement Priority Levels***

| **Requirement ID** | **Priority Level** | **Justification** |
|---|---|---|
| FR-01 | High | Respiratory monitoring is a core purpose of the component |
| FR-02 | High | Temperature monitoring is essential to identify thermal abnormalities |
| FR-03 | High | Real-time operation depends on continuous acquisition |
| FR-04 | High | Reliable detection requires cleaned and validated data |
| FR-05 | Medium | Personalized interpretation improves detection quality |
| FR-06 | High | Anomaly detection is a central function of the system |
| FR-07 | High | Risk classification supports meaningful warning decisions |
| FR-08 | High | The system must respond immediately to unsafe conditions |
| FR-09 | High | Elderly users need direct on-device warning support |
| FR-10 | Medium | External alerting strengthens caregiver awareness |
| FR-11 | High | Continuous monitoring is necessary for early warning |
| FR-12 | Medium | Integration with the overall team system supports scalability and collaboration |

## 5.3 Requirement Description Summary

***Table 5.5: Summary of User-Centred Requirement Mapping***

| **User Need** | **Related Functional Requirements** |
|---|---|
| Simple and continuous monitoring for elderly users | FR-01, FR-02, FR-03, FR-11 |
| Reliable interpretation of physiological conditions | FR-04, FR-05, FR-06, FR-07 |
| Immediate warning support in emergencies | FR-08, FR-09 |
| Remote awareness of caregivers and family | FR-10 |
| Integration with the wider healthcare platform | FR-12 |

These requirements establish the functional foundation of the Respiratory and Thermal Warning System and ensure that the component supports continuous sensing, local intelligence, timely warning, and coordination with the wider team solution.

## 5.4 Expected Test Cases

*Table: Expected Test Cases for the Respiratory and Thermal Warning System*

| **Test Case ID** | **Test Case Name** | **Objective** | **Input / Condition** | **Expected Result** | **Status** |
|---|---|---|---|---|---|
| TC-01 | $SpO_2$ Sensor Reading Validation | Verify that the system can capture oxygen saturation correctly | User wears device properly and sensor is activated | System displays and records $SpO_2$ value successfully | Expected |
| TC-02 | Heart Rate Reading Validation | Verify that heart rate is measured correctly | User wears device under normal resting condition | System displays and records heart rate value successfully | Expected |

| | | | | | |
|---|---|---|---|---|---|
| TC-03 | Temperature Sensor Reading Validation | Verify that the system can capture skin/body temperature | Temperature sensor is placed correctly on user | System displays and records temperature value successfully | Expected |
| TC-04 | Continuous Data Acquisition | Check whether the device collects readings continuously | Device runs for a defined monitoring period | Sensor values are collected continuously without interruption | Expected |
| TC-05 | Data Preprocessing Accuracy | Verify removal of invalid or noisy readings | Introduce unstable or noisy sensor input | System filters invalid values and stabilizes data before analysis | Expected |
| TC-06 | Personalized Baseline Creation | Verify that the system can establish normal baseline ranges | User wears device during a normal observation period | System stores baseline $SpO_2$, heart rate, and temperature ranges | Expected |
| TC-07 | Normal Condition Classification | Check whether normal readings are classified correctly | Input values remain within safe baseline range | System identifies condition as normal and does not trigger alert | Expected |
| TC-08 | Low $SpO_2$ Detection | Verify abnormal respiratory warning detection | Simulated or test $SpO_2$ value drops below threshold | System detects abnormal oxygen level and generates warning | Expected |
| TC-09 | Elevated Temperature Detection | Verify thermal abnormality detection | Simulated or test temperature rises above threshold | System detects abnormal temperature and generates warning | Expected |

| | | | | | |
|---|---|---|---|---|---|
| TC-10 | Combined Abnormality Detection | Check detection when multiple abnormal values occur together | Low $SpO_2$ with abnormal temperature and pulse change | System classifies higher-risk condition and triggers warning immediately | Expected |
| TC-11 | Mild Deviation Detection | Verify caution-level detection for slight abnormal variation | Slight deviation from personalized baseline | System generates caution status instead of critical warning | Expected |
| TC-12 | Real-Time Alert Generation | Verify instant alert response | Abnormal condition is detected by analysis engine | System activates local alert without major delay | Expected |
| TC-13 | Local Alert Mechanism Test | Verify wearable notification output | Warning condition is triggered | Buzzer / vibration / LED alert is activated correctly | Expected |
| TC-14 | Mobile App / Dashboard Notification | Verify communication to external monitoring interface | Warning condition is triggered and communication is active | Alert message is sent to connected mobile app or dashboard | Expected |
| TC-15 | Team System Integration Test | Verify communication with the main team platform | Warning data is generated in wearable module | Data/status is successfully transmitted to overall system | Expected |
| TC-16 | Sensor Failure Handling | Check system response to sensor disconnection or failure | Sensor is disconnected or gives null values | System identifies sensor issue and avoids false warning output | Expected |

| | | | | | |
|---|---|---|---|---|---|
| TC-17 | Power Stability Test | Verify device operation under battery-powered use | Device runs continuously on battery | System remains functional for the expected monitoring duration | Expected |
| TC-18 | Response Time Test | Measure the time taken to generate alerts | Abnormal reading is introduced | System generates warning within acceptable response time | Expected |
| TC-19 | False Alarm Reduction Test | Check whether normal fluctuations trigger unnecessary alerts | Slight temporary fluctuation in readings | System avoids false alert and maintains stable classification | Expected |
| TC-20 | End-to-End Functional Test | Validate full workflow of the system | Normal-to-abnormal physiological scenario is simulated | System completes sensing, preprocessing, detection, alerting, and communication correctly | Expected |

# 6. Commercialization Plan

The Respiratory and Thermal Warning System proposed has a clear commercialization potential being a low-priced wearable healthcare device, used in the continuous monitoring of elderly people. The focus market will be the expanding aged care industry, especially home-based care settings where the elderly need routine physiological attention but are not exposed to full time clinical attention. This is in local and international markets, where ages are driving high demand of affordable remote health monitoring devices [2], [10], [11]. The system is particularly applicable to the environment where families must look after elderly members, at home, and require a convenient early-warning device of respiratory and thermal anomalies.

The key target customers of this solution include aged people, family caregivers, private care providers, assisted living facilities and community care facilities. Continuous monitoring and timely alerts would also benefit elderly users and enable caregivers to be more reassured and become aware of potential deterioration in a much shorter time. Such a system can also be used by healthcare providers as an additional home-monitoring tool to facilitate preventive care and minimize unneeded visits to hospitals [1], [10], [12]. By doing so, the solution benefits the direct users as well as the institutional buyers.

The most fundamental value proposition of the product is the fact that it can offer real-time early warning of respiratory and thermal risk factors via a small, wearable, privacy-conscious edge-based device. In contrast to most traditional systems that require a strong reliance on cloud-based analysis or a single-parameter monitor, the suggested product consists of oxygen saturation, heart rate and temperature-based monitoring functionality in a single integrated wearable platform [3], [10]. This enhances practical utility as well as facilitating quicker reaction, reduced information exposure, and more appropriate or ageing care deployment.

Potential pricing may be based on a three-step approach. First, the wearable device can be sold as a single hardware product at the middle-range price that is affordable and can be bought by households and care-centers. Second, optional connected mobile monitoring, caregiver dashboards, alert history, and future advanced analytics models can be established. This brings flexibility to both the individual users who are price-sensitive, and institutional customers.

The system has a competitive edge in the form of low-cost architecture, elderly oriented design, multi-parameter warning and embedded intelligence. These characteristics make it stand out among simple fitness bracelets and other pricier medical trackers [1], [3], [10]. The revenue can hence be earned by selling devices, institutional packages, mobile application services, and future licensing opportunities. The system can result in patentable wearable design, embedded detection structure, or integrated alert system in terms of intellectual property and future product potential. The product may also be extended to cover a wider smart elderly healthcare platform with more sensing and tele-health functionality in the future [10], [11].

# 7. Budget and Justification

***Table 7.1: Estimated Budget and Justification for the Respiratory and Thermal Warning System***

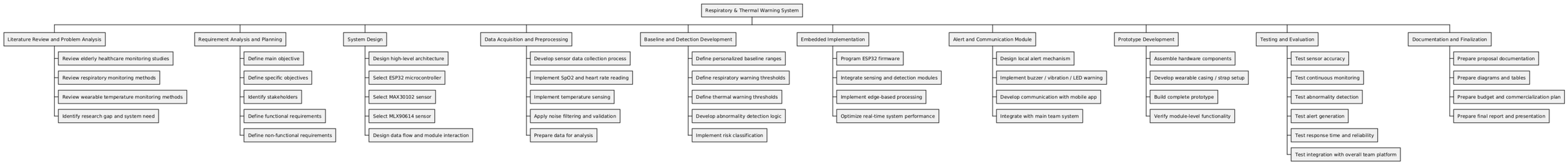
| Budget Category | Item | Quantity | Estimated Unit Cost (LKR) | Estimated Total Cost (LKR) | Justification |
|---|---|---|---|---|---|
| Hardware | ESP32 Microcontroller | 1 | 3,500 | 3,500 | Required as the main embedded processing platform for sensor integration, local analysis, and communication. |
| Hardware | MAX30102 Sensor Module | 1 | 2,000 | 2,000 | Needed to measure blood oxygen saturation and heart rate for respiratory-related monitoring. |

| | | | | | |
|---|---|---|---|---|---|
| Hardware | MLX90614 Temperature Sensor | 1 | 4,500 | 4,500 | Required to monitor temperature abnormalities as part of the thermal warning function. |
| Hardware | Buzzer / Vibration Motor / LED Alert Components | 1 set | 1,500 | 1,500 | Needed to generate immediate local alerts when abnormal conditions are detected. |
| Hardware | Breadboard / PCB / Prototyping Board | 1 | 1,500 | 1,500 | Used for circuit assembly, module integration, and prototype development. |
| Hardware | Jumper Wires / Connectors / Resistors / Basic Electronics | 1 set | 1,500 | 1,500 | Necessary for hardware interconnection and stable electronic setup. |
| Hardware | Rechargeable Battery and Power Module | 1 | 3,000 | 3,000 | Supports portable wearable operation and continuous monitoring without fixed power supply. |
| Hardware | Wearable Strap / Enclosure / Casing Materials | 1 | 2,500 | 2,500 | Required to convert the prototype into a wearable form suitable for elderly users. |

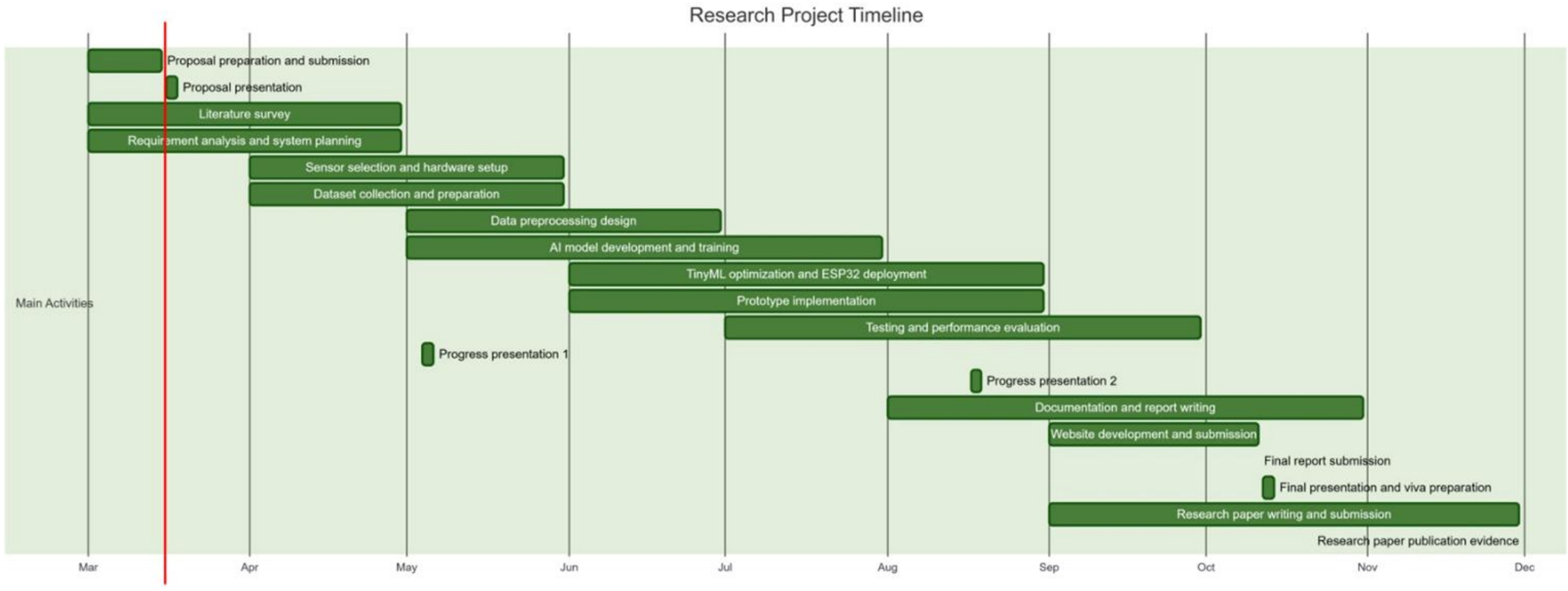
| | | | | | |
|---|---|---|---|---|---|
| Hardware | Testing and Replacement Components | 1 set | 2,000 | 2,000 | Covers damaged parts, repeated testing needs, and minor hardware replacement during development. |
| Software / Tools | Arduino IDE / Embedded Development Tools | 1 | 0 | 0 | Open-source development tools used for firmware programming and hardware integration. |
| Software / Tools | Data Analysis / Model Testing Tools | 1 | 0 | 0 | Open-source tools are sufficient for signal analysis, validation, and prototype testing. |
| Mobile / Integration | Mobile App / Dashboard Prototype Integration | 1 | 5,000 | 5,000 | Estimated cost for basic interface development, connectivity testing, and alert display integration. |
| Mobile / Integration | Cloud / Hosting / API Testing Allowance | 1 | 3,000 | 3,000 | Small allocation for optional testing of notification handling, storage, or remote connectivity support. |

| | | | | | |
|---|---|---|---|---|---|
| Human Effort | Prototype Design and Hardware Assembly | 1 | 8,000 | 8,000 | Represents the effort involved in sensor setup, wearable assembly, and integration work. |
| Human Effort | Software Development and Embedded Programming | 1 | 10,000 | 10,000 | Covers firmware development, analysis logic implementation, and alerting functions. |
| Human Effort | Testing, Calibration, and Validation | 1 | 7,000 | 7,000 | Required for repeated testing, adjustment of thresholds, and evaluation of detection behaviour. |
| Contingency | Miscellaneous / Unforeseen Expenses | 1 | 4,000 | 4,000 | Reserved for unexpected prototype adjustments, minor purchases, or additional testing needs. |
| | | | **Grand Total** | **64,000** | |

## 8. Work Breakdown Structure (WBS) diagram




## 9. GANTT CHART



# 10. References List

[1] D. Dias and J. P. S. Cunha, “Wearable Health Devices—Vital Sign Monitoring, Systems and Technologies,” *Sensors*, vol. 18, no. 8, p. 2414, 2018. doi: 10.3390/s18082414. Available: https://doi.org/10.3390/s18082414

[2] L. Lu, J. Zhang, Y. Xie, F. Gao, S. Xu, X. Wu, and Z. Ye, “Wearable Health Devices in Health Care: Narrative Systematic Review,” *JMIR mHealth and uHealth*, vol. 8, no. 11, p. e18907, 2020. doi: 10.2196/18907. Available: https://doi.org/10.2196/18907

[3] P. P. Ray, “A Review on TinyML: State-of-the-Art and Prospects,” *Journal of King Saud University - Computer and Information Sciences*, vol. 34, pp. 1595–1623, 2022. doi: 10.1016/j.jksuci.2021.11.019. Available: https://doi.org/10.1016/j.jksuci.2021.11.019

[4] J. S. Sunny, C. P. K. Patro, K. Karnani, S. C. Pingle, F. Lin, M. Anekoji, L. D. Jones, S. Kesari, and S. Ashili, “Anomaly Detection Framework for Wearables Data: A Perspective Review on Data Concepts, Data Analysis Algorithms and Prospects,” *Sensors*, vol. 22, no. 3, p. 756, 2022. doi: 10.3390/s22030756. Available: https://doi.org/10.3390/s22030756

[5] T. Hussain, S. Ullah, R. Fernández-García, and I. Gil, “Wearable Sensors for Respiration Monitoring: A Review,” *Sensors*, vol. 23, no. 17, p. 7518, 2023. doi: 10.3390/s23177518. Available: https://doi.org/10.3390/s23177518

[6] P. H. Charlton, D. Birrenkott, T. Bonnici, M. A. F. Pimentel, A. E. W. Johnson, J. Alastruey, L. Tarassenko, P. J. Watkinson, R. J. Beale, and D. A. Clifton, “Breathing Rate Estimation From the Electrocardiogram and Photoplethysmogram: A Review,” *IEEE Reviews in Biomedical Engineering*, vol. 11, pp. 2–20, 2018. doi: 10.1109/RBME.2017.2763681. Available: https://doi.org/10.1109/RBME.2017.2763681

[7] E. D. Chan, M. M. Chan, and M. M. Chan, “Pulse Oximetry: Understanding Its Basic Principles Facilitates Appreciation of Its Limitations,” *Respiratory Medicine*, vol. 107, no. 6, pp. 789–799, 2013. doi: 10.1016/j.rmed.2013.02.004. Available: https://doi.org/10.1016/j.rmed.2013.02.004

[8] C. Shi, M. Goodall, J. Dumville, J. Hill, G. Norman, O. Hamer, A. Clegg, C. L. Watkins, G. Georgiou, A. Hodkinson, C. E. Lightbody, P. Dark, and N. Cullum, “The Accuracy of Pulse Oximetry in Measuring Oxygen Saturation by Levels of Skin Pigmentation: A

Systematic Review and Meta-Analysis," *BMC Medicine*, vol. 20, no. 1, p. 267, 2022. doi: 10.1186/s12916-022-02452-8. Available: https://doi.org/10.1186/s12916-022-02452-8

[9] A. M. Luks and E. R. Swenson, "Pulse Oximetry for Monitoring Patients with COVID-19 at Home: Potential Pitfalls and Practical Guidance," *Annals of the American Thoracic Society*, vol. 17, no. 9, pp. 1040–1046, 2020. doi: 10.1513/AnnalsATS.202005-418FR. Available: https://doi.org/10.1513/AnnalsATS.202005-418FR

[10] W. Jiang, S. Majumder, S. Kumar, S. Subramaniam, X. Li, R. Khedri, T. Mondal, M. Abolghasemian, I. Satia, and M. J. Deen, "A Wearable Tele-Health System towards Monitoring COVID-19 and Chronic Diseases," *IEEE Reviews in Biomedical Engineering*, vol. 15, pp. 61–84, 2022. doi: 10.1109/RBME.2021.3069815. Available: https://doi.org/10.1109/RBME.2021.3069815

[11] Imran, N. Iqbal, S. Ahmad, and D.-H. Kim, "Health Monitoring System for Elderly Patients Using Intelligent Task Mapping Mechanism in Closed Loop Healthcare Environment," *Symmetry*, vol. 13, no. 2, p. 357, 2021. doi: 10.3390/sym13020357. Available: https://doi.org/10.3390/sym13020357

[12] A. F. Silva and M. Tavakoli, "Domiciliary Hospitalization through Wearable Biomonitoring Patches: Recent Advances, Technical Challenges, and the Relation to Covid-19," *Sensors*, vol. 20, no. 23, p. 6835, 2020. doi: 10.3390/s20236835. Available: https://doi.org/10.3390/s20236835

testing - dissanayaka.pdf

ORIGINALITY REPORT

| 1% | 1% | 0% | 0% |
|---|---|---|---|
| SIMILARITY INDEX | INTERNET SOURCES | PUBLICATIONS | STUDENT PAPERS |

PRIMARY SOURCES

| | | |
|---|---|---|
| 1 | www.psc.gov.bd<br>Internet Source | <1% |
| 2 | sci-hub.se<br>Internet Source | <1% |
| 3 | Submitted to Middle East College<br>Student Paper | <1% |
| 4 | shodhgangotri.inflibnet.ac.in<br>Internet Source | <1% |
| | e-archivo.uc3m.es | |



## *% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

**Caution: Review required.**

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

### Detection Groups

**0** AI-generated only 0%
Likely AI-generated text from a large-language model.

**0** AI-generated text that was AI-paraphrased 0%
Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

**Disclaimer**
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

### Frequently Asked Questions

**How should I interpret Turnitin's AI writing percentage and false positives?**
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.